COMP1017 (Mathematics for Computer Scientists) Test 2

Academic Session 2017-2018, Autumn Semester

There are TWO (2) questions pertaining to sets and functions on ONE (1) page. Please answer ALL the questions. Time allowed 30 minutes.

1. There are 400 students in a college. Each of the students is required to read several different newspapers. Let S denote the set of students in the college, and T the set of newspapers that are provided to all the students, respectively.

(a) Determine $|S|, |P(S)|$, and $|S \times T|$. Show your work (you do not have to give the final figures).

$|S| = 400$, $|P(S)| = 2^{400}$, $|S \times T|$ is not determined [15 marks]

Since we don't really know the cardinality of T.

(b) Assume that each newspaper is read by exactly 80 students. Let A and B represent the set of students who have read the newspapers 'a' and 'b', respectively. Determine the range of $|A \cup B|$. Show your work.

$|A \cup B| = |A| + |B| - |A \cap B|$, Since $0 \leq |A \cap B| \leq 80$ [16 marks]

thus $80 \leq |A \cup B| \leq 160$

(c) Assume that there are 8 different newspapers in total. Define a function $f: S \to P(T)$ such that $f(x) = t$, where the cardinality $|t| = 5$. Is f injective? Is f surjective? Explain your answers.

Note that $|S| = 400$. $|P(T)| = 2^8 = 256$, therefore [19 marks]

f cannot be injective no matter how it's defined.

f can't be surjective, Since the range $= \{t \in P(T) \mid |t| = 5\} \subset P(T)$.

2. In the questions below, let both f and g be functions.

(a) Define $f, g: R \to Z$, such that $f = \lfloor x^2 \rfloor$, and $g = \left\lceil \frac{x+1}{2} \right\rceil$. Determine $f \circ g(-1.5)$ and $g \circ f(-1.5)$, respectively. Show your work.

$f \circ g(-1.5) = f(g(-1.5)) = \left\lfloor \left\lceil \frac{(-1.5)+1}{2} \right\rceil^2 \right\rfloor = \lfloor 0^2 \rfloor = 0$ [16 marks]

$g \circ f(-1.5) = g(f(-1.5)) = \left\lceil \frac{\lfloor (-1.5)^2 \rfloor + 1}{2} \right\rceil = \left\lceil \frac{2+1}{2} \right\rceil = 2$

(b) Let $A = \{1, 2, 3, 4\}$ and $B = \{a, b, c, d\}$. Define $f: A \to B$ and $g: B \to A$ such that $f(1) = a, f(2) = d, f(3) = c, f(4) = b$, and $g(a) = 3, g(b) = 2, g(c) = 4, g(d) = 1$, respectively. Determine $(f \circ g)^{-1}$. Show your work.

$f \circ g(a) = f(g(a)) = f(3) = c$, $f \circ g(b) = f(g(b)) = d$. $f \circ g(c) = f(g(c)) = b$ [16 marks]

$f \circ g(d) = f(g(d)) = a$, Thus $(f \circ g)^{-1}(a) = d$. $(f \circ g)^{-1}(b) = c$, $(f \circ g)^{-1}(c) = a$, $(f \circ g)^{-1}(d) = b$

(c) The sets A and B are defined as in (b). Let B_1 and B_2 be two subsets of B respectively. Define $S = \{f \mid f: A \to B_1\}$ the set of all functions mapping A to B_1, and $T = \{g \mid g: A \to B_2\}$ the set of all functions mapping A to B_2. Show that if $B_1 \subseteq B_2$ then $S \subseteq T$.

$\forall f \in S$, $\forall a \in A$, $\exists b \in B_1$ such that $f(a) = b$ [18 marks]

Since $B_1 \subseteq B_2$, we have $b \in B_2$. this means $\forall a \in A$, $\exists b \in B_2$ such that $f(a) = b$, That is to say $f \in T$.

Thus it is shown $S \subseteq T$.